Module 7

# Capital Budgeting

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

# Goals

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# Goals

- How to decide what project(s) to do? The tool is the NPV rule
- But to use NPV, we need a project's cash flows
- Here, combine what we know about accounting and cash flow effects (**Module 2**) with the NPV rule (**Module 6**), to decide whether or not to accept a project
- **Capital budgeting**, the process of analyzing capital investment opportunities ("projects") and deciding which ones to accept

# Roadmap

## Should we invest in a project (e.g., a data center)?

**NPV $> 0$?**

Recall from **Module 6**

# Roadmap

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## Should we invest in a project (e.g., a data center)?

**How to forecast cash flows?**

- Similar to **Module 2**, indirect method for cash flow
- But different, cash flow from a project, not a whole firm

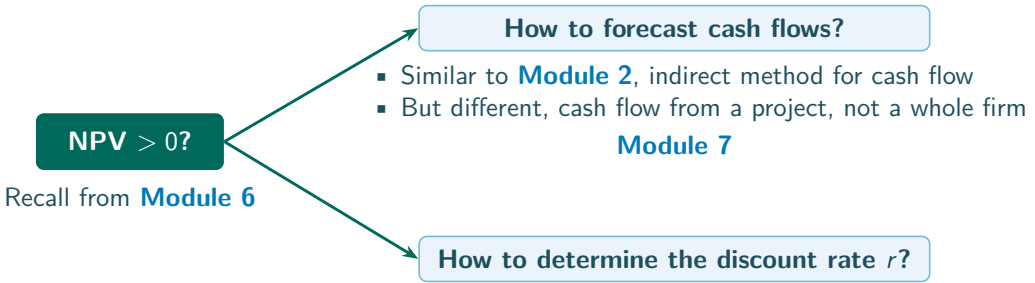
**Module 7**

**NPV $> 0$?**

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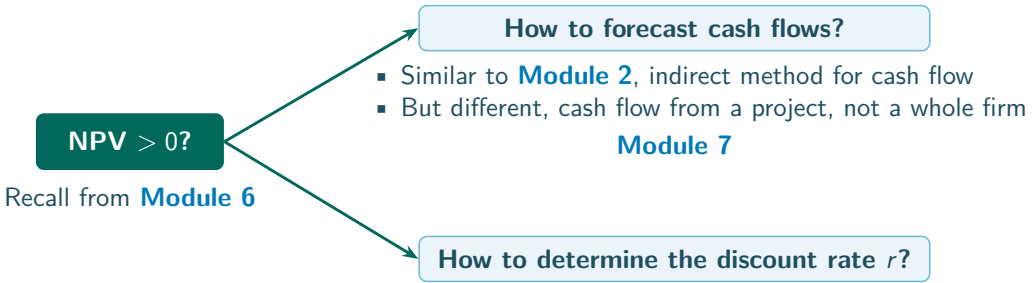
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# Motivating Example, Is AI Spending a Bubble?




CNBC, August 12, 2026 (7:21), "We're 'absolutely' in an AI bubble right now, says Lead Edge Capital's Mitchell Green"

## Module 7, Part 0

# Review of the Statement of Cash Flows

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# Statement of Cash Flows

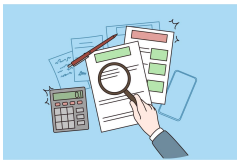
Recall from **Module 2**

| Direct method | Indirect method |
|---|---|
| Lists every cash receipt and payment | Starts from net income and adjusts it |
| More time-consuming to prepare | Faster to prepare |
| Rarely used in practice | **The common choice** |

**Direct** →



**Indirect** →



# Statement of Cash Flows

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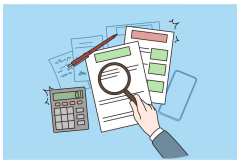
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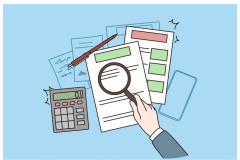
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

**Direct** →



**Indirect** →



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***Relationship of cash flow (CF) and net income (NI)?***
***When spending on fixed assets, CF = NI − spending on fixed assets***
***When recording depreciation, CF = NI + depreciation***

# How Do Net Income and Cash Flow Differ?

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- The firm spends $60 million producing AI control panels
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OpenAI, July 15, 2026 (2:15),
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Recall from **Module 2**

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***When firm pays off suppliers, CF = NI + change in accounts payable***

# Operating Activities

Recall from **Module 2**

**Operating activities**, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
   - NWC = inventories + accounts receivable − accounts payable
   - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

# Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

***When adding to inventories, CF = NI − change in inventories***

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$$CF = NI - \text{change in net working capital (NWC)}$$

# Investing Activities

Recall from **Module 2**

**Investing activities**, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
   - **Capital**, a firm's fixed assets
   - **Capital expenditure**, the amount spent on buying capital during the period
   - Cash flow is *negative* if firms buy capital

# Financing Activities

Recall from **Module 2**

**Financing activities**, cash raised from lenders and shareholders, and cash returned to them

# Recap of Part 0

Recall from **Module 2**

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
  - **Operating activities**, cash flow = Net income + Depreciation − Change in NWC
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
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## Module 7, Part 1

# Project Cash Flow

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# Firm Cash Flow vs. Project Cash Flow

- **Module 2** looked at the whole firm, here we look at one project on its own
- **Operating activities**, the same idea, project net income plus depreciation minus the change in NWC
- **Investing activities**, the project's capital investments, buying its assets at the start
- **Financing activities**, *none*, we do not consider financing activities when evaluating a project on its own

# Identify Project Cash Flow

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$$\text{Cash flow from a project} = \begin{cases} \text{Project net income} & \text{Component 1} \\ + \text{ Depreciation} & \text{Component 2} \\ \text{Cash flow from changes in NWC} & \text{Component 3} \\ \text{Cash flow from capital investments} & \text{Component 4} \end{cases}$$

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| |
|---|
| Revenues from the project |
| – COGS and SG&A expenses for the project |
| – Depreciation for the project's capital investment |
| **Project EBIT** |
| – Taxes (tax rate $\times$ project EBIT) |
| **Project net income** |

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- To get cash flow from the project, add back depreciation, as in **Module 2**

# Example, Project Net Income and Depreciation

- A project requires an investment in fixed assets of $2,000. The project's revenues minus its COGS and SG&A expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%

# Example, Project Net Income and Depreciation

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- Calculate the project's net income plus depreciation for year 1

# Example, Project Net Income and Depreciation

- Solution,

| Item | $ |
|---|---|
| Revenues − COGS − SG&A expenses | 1,100 |
| − Depreciation | 1,000 |
| **Project EBIT** | **100** |
| − Taxes | 21 |
| **Project net income** | **79** |
| + Depreciation | 1,000 |
| **Project net income + Depreciation** | **1,079** |

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- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$
- Project net income + Depreciation $= 79 + 1{,}000 = \$1{,}079$

# Component 3, Cash Flow from Changes in NWC

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- A project has its own NWC, e.g.,
  - an inventory of raw materials before production starts
  - accounts receivable rising as sales start
  - accounts payable rising as the project buys supplies on credit
- An increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow,

***Cash flow from changes in NWC = −Change in the project's NWC***

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| Year | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| The project's NWC | 100 | 100 | 100 | 0 |
| Change in NWC | $+100$ | 0 | 0 | $-100$ |
| Cash flow from changes in NWC | $-100$ | 0 | 0 | $+100$ |

# Component 3, Cash Flow from Changes in NWC

- E.g., a 3-year project needs \$100 of NWC while it operates, recovered when the project ends

| Year | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| The project's NWC | 100 | 100 | 100 | 0 |
| Change in NWC | $+100$ | 0 | 0 | $-100$ |
| Cash flow from changes in NWC | $-100$ | 0 | 0 | $+100$ |

- The \$100 goes out at $t = 0$ and returns only at $t = 3$, so tying up the cash still costs the time value of money

# Component 4, Cash Flow from Capital Investments

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# Component 4, Cash Flow from Capital Investments

- If the project requires its own investment in fixed assets to get started (e.g., building a data center), that is a negative cash flow
- Fixed assets are usually depreciated "straight-line" over time,
  - **Expected life**, the number of years the project uses the fixed assets

$$\text{Annual depreciation} = \frac{\text{Capital investment}}{\text{Expected life}}$$

# Identify Project Cash Flow

- Cash flow from a project has four components,

$$\text{Cash flow from a project} = \begin{cases} \text{Project net income} & \text{Component 1} \\ + \text{ Depreciation} & \text{Component 2} \\ \text{Cash flow from changes in NWC} & \text{Component 3} \\ \text{Cash flow from capital investments} & \text{Component 4} \end{cases}$$

# Recap of Part 1

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- The investment in fixed assets is a negative cash flow, and the fixed assets are depreciated straight-line over the expected life

## Module 7, Part 2

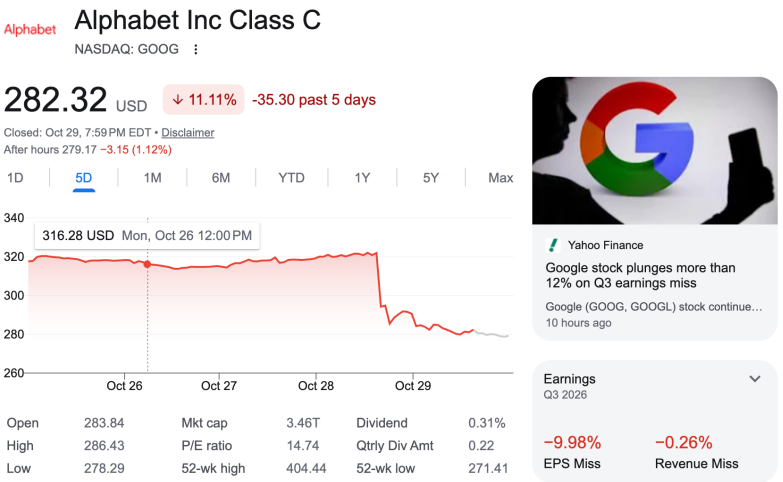
# Putting It All Together

# Motivating Example, Google's AI Spend under Scrutiny




CNBC, October 28, 2026 (3:06), "Google's AI spend under scrutiny, here's what to know"

# Motivating Example, Google's AI Spend under Scrutiny



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- A company is considering building a new data center, an investment in fixed assets of $1 billion ($1,000 million) that will last 5 years. It will be depreciated straight-line over 5 years

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- Renting out its computing capacity brings in \$240 million a year after COGS and SG&A expenses (electricity, staff, and so on)
- The corporate tax rate is 21%, and the discount rate is 8%
- Should the company build the data center or not?

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# Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |
| | | | | | | |

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- Build the project's income statement, the same in years 1 to 5 ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 240 | 240 | 240 | 240 | 240 |
| | | | | | | |
| | | | | | | |

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- Depreciation $= 1{,}000/5 = 200$ a year

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| **Project EBIT** | | **40** | **40** | **40** | **40** | **40** |
| − Taxes | | 8.4 | 8.4 | 8.4 | 8.4 | 8.4 |

- Depreciation $= 1{,}000/5 = 200$ a year
- Taxes $= 21\% \times 40 = 8.4$

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| Revenues − COGS − SG&A expenses | | 240 | 240 | 240 | 240 | 240 |
| − Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project EBIT** | | **40** | **40** | **40** | **40** | **40** |
| − Taxes | | 8.4 | 8.4 | 8.4 | 8.4 | 8.4 |
| **Project net income** | | **31.6** | **31.6** | **31.6** | **31.6** | **31.6** |

- Depreciation $= 1{,}000/5 = 200$ a year
- Taxes $= 21\% \times 40 = 8.4$

# Example, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ millions),

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|---|---|---|---|---|---|---|
| | | | | | | |

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|---|---|---|---|---|---|---|
| Project net income | | 31.6 | 31.6 | 31.6 | 31.6 | 31.6 |
| + Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project net income + Depreciation** | | **231.6** | **231.6** | **231.6** | **231.6** | **231.6** |

- The cash for the data center went out at $t = 0$, it is counted in Component 4

# Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|

# Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| The project's NWC | 100 | 100 | 100 | 100 | 100 | 0 |

# Example, Component 3, Cash Flow from Changes in NWC

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| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| The project's NWC | 100 | 100 | 100 | 100 | 100 | 0 |
| Change in NWC | +100 | 0 | 0 | 0 | 0 | −100 |

# Example, Component 3, Cash Flow from Changes in NWC

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| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| The project's NWC | 100 | 100 | 100 | 100 | 100 | 0 |
| Change in NWC | +100 | 0 | 0 | 0 | 0 | −100 |
| **Cash flow from changes in NWC** | **−100** | **0** | **0** | **0** | **0** | **+100** |

- An increase in NWC is a negative cash flow, and a decrease is a positive cash flow

# Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|

# Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Investment in fixed assets | −1,000 | 0 | 0 | 0 | 0 | 0 |

# Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Investment in fixed assets | −1,000 | 0 | 0 | 0 | 0 | 0 |
| **Cash flow from capital investments** | **−1,000** | **0** | **0** | **0** | **0** | **0** |

# Example, Total Cash Flows

- Add up the four components year by year ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|

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| Project net income + Depreciation | 0 | 231.6 | 231.6 | 231.6 | 231.6 | 231.6 |
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# Example, Total Cash Flows

- Add up the four components year by year ($ millions),

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|---|---|---|---|---|---|---|
| Project net income + Depreciation | 0 | 231.6 | 231.6 | 231.6 | 231.6 | 231.6 |
| Cash flow from changes in NWC | −100 | 0 | 0 | 0 | 0 | +100 |
| Cash flow from capital investments | −1,000 | 0 | 0 | 0 | 0 | 0 |
| **Total cash flows** | **−1,100** | **231.6** | **231.6** | **231.6** | **231.6** | **331.6** |

- At $t = 0$, $-1{,}000 - 100 = -1{,}100$. At $t = 5$, $231.6 + 100 = 331.6$

# Example, NPV and the Decision

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$$\text{NPV} = -1{,}100 + \frac{231.6}{1.08} + \frac{231.6}{1.08^2} + \frac{231.6}{1.08^3} + \frac{231.6}{1.08^4} + \frac{331.6}{1.08^5} = -107.23$$

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- NPV $< 0$, so the company should not build the data center

# Example, NPV with the Cash Flow Worksheet

**One cash flow at a time**

| Enter | Keys |
|---|---|
| | **CF**, **2nd**, **CLR Work** |
| −1100 | **ENTER**, ↓ |
| 231.6 | **ENTER**, ↓ ↓ |
| 231.6 | **ENTER**, ↓ ↓ |
| 231.6 | **ENTER**, ↓ ↓ |
| 231.6 | **ENTER**, ↓ ↓ |
| 331.6 | **ENTER** |
| | **NPV** (the calculator asks for I) |
| 8 | **ENTER**, ↓ |
| | **CPT** |
| **−107.23** | |

# Example, NPV with the Cash Flow Worksheet

**One cash flow at a time**

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| 331.6 | ENTER |
| | NPV (the calculator asks for I) |
| 8 | ENTER, ↓ |
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**Repeated cash flow with frequency**

| Enter | Keys |
|---|---|
| | CF, 2nd, CLR Work |
| −1100 | ENTER, ↓ |
| 231.6 | ENTER, ↓ |
| 4 | ENTER (the frequency F01), ↓ |
| 331.6 | ENTER |
| | NPV (the calculator asks for I) |
| 8 | ENTER, ↓ |
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# Example, IRR with the Cash Flow Worksheet

- Recall from **Module 6**, the IRR is the discount rate at which NPV $= 0$

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|---|---|
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| $-1100$ | **ENTER**, ↓ |
| 231.6 | **ENTER**, ↓ |
| 4 | **ENTER** (the frequency F01), ↓ |
| 331.6 | **ENTER** |
| | **IRR** (shows 0) |
| | **CPT** |
| **4.41%** | |

# Example, IRR with the Cash Flow Worksheet

- Recall from **Module 6**, the IRR is the discount rate at which NPV $= 0$
- The same cash flow worksheet, then **IRR**, **CPT**
- IRR (4.41%) $<$ discount rate (8%), reject the project, the same answer as the NPV rule

| Enter | Keys |
|---|---|
| | **CF**, **2nd**, **CLR Work** |
| $-1100$ | **ENTER**, ↓ |
| 231.6 | **ENTER**, ↓ |
| 4 | **ENTER** (the frequency F01), ↓ |
| 331.6 | **ENTER** |
| | **IRR** (shows 0) |
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3. Add the cash flow from capital investments, the investment in fixed assets at the start and any terminal cash flows at the end (Component 4)
4. Total the cash flows year by year, discount at the discount rate, and apply the NPV rule

## Module 7, Part 3

# The Principles behind Project Cash Flows

# General Principles for Identifying Project Cash Flow

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3. Consider all and only incremental cash flows

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# Principle 1, Discount Cash Flows, Not Net Income

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- Differences in how capital investments, depreciation, and NWC are handled
- We always discount the actual cash flows to their present value
- We do not discount net income

# Example, Net Income vs. Cash Flows

- A project requires an investment in fixed assets of $2,000 and lasts 2 years. The fixed assets are depreciated straight-line at $1,000 per year

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- The discount rate is 10%

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- The discount rate is 10%
- Compare the NPV using net income to the NPV using cash flows

***Clearly a money-loser!***

# Example, Net Income vs. Cash Flows

- Solution,

| Year | 0 | 1 | 2 |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

# Example, Net Income vs. Cash Flows

- Solution,

| Year | 0 | 1 | 2 |
|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 1,500 | 500 |
| − Depreciation | | 1,000 | 1,000 |

# Example, Net Income vs. Cash Flows

- Solution,

| Year | 0 | 1 | 2 |
|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 1,500 | 500 |
| − Depreciation | | 1,000 | 1,000 |
| **Project net income** | | **500** | **−500** |

# Example, Net Income vs. Cash Flows

- Solution,

| Year | 0 | 1 | 2 |
|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 1,500 | 500 |
| − Depreciation | | 1,000 | 1,000 |
| **Project net income** | | **500** | **−500** |
| + Depreciation | | 1,000 | 1,000 |
| **Project net income + Depreciation** | **0** | **1,500** | **500** |

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|---|---|---|---|
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| **Project net income + Depreciation** | **0** | **1,500** | **500** |
| Cash flow from capital investments | −2,000 | 0 | 0 |

# Example, Net Income vs. Cash Flows

- Solution,

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|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 1,500 | 500 |
| − Depreciation | | 1,000 | 1,000 |
| **Project net income** | | **500** | **−500** |
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| **Project net income + Depreciation** | **0** | **1,500** | **500** |
| Cash flow from capital investments | −2,000 | 0 | 0 |
| **Total cash flows** | **−2,000** | **1,500** | **500** |

# Example, Net Income vs. Cash Flows

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| Year | 0 | 1 | 2 |
|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 1,500 | 500 |
| − Depreciation | | 1,000 | 1,000 |
| **Project net income** | | **500** | **−500** |
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- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project

# Example, Net Income vs. Cash Flows

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|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 1,500 | 500 |
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| **Project net income + Depreciation** | **0** | **1,500** | **500** |
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- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project
- NPV using cash flows $= -2{,}000 + \frac{1{,}500}{1.10} + \frac{500}{1.10^2} = -\$223.14$

# Example, Net Income vs. Cash Flows

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| Year | 0 | 1 | 2 |
|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 1,500 | 500 |
| − Depreciation | | 1,000 | 1,000 |
| **Project net income** | | **500** | **−500** |
| + Depreciation | | 1,000 | 1,000 |
| **Project net income + Depreciation** | **0** | **1,500** | **500** |
| Cash flow from capital investments | −2,000 | 0 | 0 |
| **Total cash flows** | **−2,000** | **1,500** | **500** |

- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project
- NPV using cash flows $= -2{,}000 + \frac{1{,}500}{1.10} + \frac{500}{1.10^2} = -\$223.14$
- NPV $< 0$, so the project should be rejected, net income gets it wrong

# Principle 2, Exclude Cash Flows from Financing Activities

- Project cash flows exclude cash flows from financing activities

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- Project cash flows exclude cash flows from financing activities
- The mini-income statement for this data center project has no interest expense row, even if the company borrows to build it

# Principle 3, Consider All and Only Incremental Cash Flows

- We need to discount **incremental** cash flows
- What does "incremental" mean in capital budgeting?

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- We need to discount **incremental** cash flows
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- Ask yourself, "would the cash flow still exist if the project did not exist?"
  - If yes, do *not* include it in the analysis
  - If no, include it

# Principle 3, Consider All and Only Incremental Cash Flows

- We need to discount **incremental** cash flows
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- Ask yourself, "would the cash flow still exist if the project did not exist?"
  - If yes, do *not* include it in the analysis
  - If no, include it

***Incremental cash flow = Cash flow with the project − Cash flow without it***

# Incremental Cash Flows, a Few Rules That Are Easy to Forget

a. Include all indirect effects

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a. Include all indirect effects
b. Exclude sunk costs
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d. Include cash flows from changes in NWC
e. Include terminal cash flows

# Rule a, Indirect Effects

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- E.g., **cannibalization**, a new product cutting into the sales of the company's existing products

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- Nintendo plans a new-generation gaming console
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- Without the new console, the old one is forecast to create $800m and $1,000m in years 1 and 2, with the new console, only $500m and $400m






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- But the launch will hurt sales of the older console
- Without the new console, the old one is forecast to create $800m and $1,000m in years 1 and 2, with the new console, only $500m and $400m
- What is the **incremental** cash flow of the new console?






# Rule a, Indirect Effects

- Solution ($ millions),

| Year | 0 | 1 | 2 |
|---|---|---|---|
| Cash flow from investment in and sales of the new one | −200 | 400 | 800 |
| Cash flow from the old console without the new one | | 800 | 1,000 |
| Cash flow from the old console with the new one | | 500 | 400 |
| **Change in cash flow from the old console** | | −300 | −600 |
| **Incremental cash flow of the new console** | −200 | $400 - 300 = 100$ | $800 - 600 = 200$ |

# Rule a, Indirect Effects

- Solution ($ millions),

| Year | 0 | 1 | 2 | |
|---|---|---|---|---|
| Cash flow from investment in and sales of the new one | −200 | 400 | 800 | ← Direct effect |
| Cash flow from the old console without the new one | | 800 | 1,000 | |
| Cash flow from the old console with the new one | | 500 | 400 | |
| **Change in cash flow from the old console** | | −300 | −600 | ← Indirect effect |
| **Incremental cash flow of the new console** | −200 | $400 - 300 = 100$ | $800 - 600 = 200$ | |

# Rule b, Sunk Costs

- **Sunk costs**, expenses that have already been paid, or expenses that can no longer be changed
  - E.g., previous R&D (already paid), rent under an unbreakable lease (can no longer be changed)



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  - E.g., previous R&D (already paid), rent under an unbreakable lease (can no longer be changed)
- A sunk cost is the same with or without the project $\rightarrow$ it is not incremental and should not affect the NPV $\rightarrow$ Exclude it!
- Many sunk costs are "related" to the project. But if the decision does not change them, they are not incremental



# Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
- Launching requires another $1 million investment (marketing and so on), with a present value of the project's future cash flows of $1.5 million

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- Proceed with the project or abort?
  - The \$0.6 million R&D is a sunk cost, the same whether you proceed or abort $\rightarrow$ Exclude it!
  - NPV $= \$1.5 - \$1 = \$0.5$ million $> 0 \rightarrow$ Proceed!

# Rule c, Opportunity Costs

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  - E.g., a piece of land the company already owns is not free, the firm could use it for another project, rent it out, or sell it

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- E.g., you plan a new factory building that costs $10m to construct, on a piece of land you already own. Alternatively, you could sell the land for $3m

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- What is the actual cost of building the factory?
  - The land is not free, its opportunity cost is the \$3m you give up by not selling it
  - Actual cost = \$10m + \$3m = \$13m

# Rule d, Changes in NWC

- If a project needs its own NWC, the cash flows from changes in NWC belong in the analysis

# Rule d, Changes in NWC

- If a project needs its own NWC, the cash flows from changes in NWC belong in the analysis
- The project's NWC often keeps changing during the project, and is often recovered at the end

# Practice Problem

- A company's project runs for 3 years. In $ thousands, its revenues are 1,000 in year 1, 2,000 in year 2, and 1,000 in year 3
- The project's NWC at each date must be 10% of the following one-year period's revenues

# Practice Problem

- A company's project runs for 3 years. In $ thousands, its revenues are 1,000 in year 1, 2,000 in year 2, and 1,000 in year 3
- The project's NWC at each date must be 10% of the following one-year period's revenues
- What are the cash flows from changes in NWC?

# Practice Problem

- Solution ($ thousands), NWC at each date is 10% of the following one-year period's revenues,

| Year | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Revenues | | 1,000 | 2,000 | 1,000 |
| The project's NWC | 100 | 200 | 100 | 0 |
| Change in NWC | +100 | +100 | −100 | −100 |
| Cash flow from changes in NWC | −100 | −100 | +100 | +100 |

# Practice Problem

- Solution ($ thousands), NWC at each date is 10% of the following one-year period's revenues,

| Year | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Revenues | | 1,000 | 2,000 | 1,000 |
| The project's NWC | 100 | 200 | 100 | 0 |
| Change in NWC | +100 | +100 | −100 | −100 |
| Cash flow from changes in NWC | −100 | −100 | +100 | +100 |

- NWC follows revenues up and down, and is fully recovered by the end of the project

# Rule e, Terminal Cash Flows

- At the end, there may be cash flows from shutting down the project, the "terminal" cash flows



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- At the end, there may be cash flows from shutting down the project, the "terminal" cash flows
- They can be positive
  - E.g., selling the project's fixed assets
- They can also be negative
  - E.g., oil wells require "capping" when they no longer produce
  - But apparently many companies still do this, Bloomberg, "It Pays to Not Pay Your Debts"



# Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant
b. The cost of a new access road put in last year
c. The market value of the land
d. The market value of the existing buildings
e. Demolition and clearance costs
f. Construction cost of the new plant
g. The initial investment in inventories of raw materials
h. Lost cash flows on other projects due to executive time spent on the new plant

# Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant ✗
b. The cost of a new access road put in last year
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# Recap of Part 3

- Discount cash flows, not net income, and exclude cash flows from financing activities
- Consider all and only incremental cash flows, "would the cash flow still exist if the project did not exist?"
- Include indirect effects, opportunity costs, changes in NWC, and terminal cash flows, exclude sunk costs

## Module 7, Part 4

# Scenario Analysis

# Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - An investment in fixed assets of 1,000 that lasts 5 years, depreciated straight-line over 5 years

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  - It brings in 240 a year after COGS and SG&A expenses

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  - The corporate tax rate is 21%, and the discount rate is 8%

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  - It needs an increase in NWC of 100 at $t = 0$, recovered fully at $t = 5$
  - It brings in 240 a year after COGS and SG&A expenses
  - The corporate tax rate is 21%, and the discount rate is 8%
- This time, build the whole table in one go

# Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |

# Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 240 | 240 | 240 | 240 | 240 |

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- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 240 | 240 | 240 | 240 | 240 |
| − Depreciation | | 200 | 200 | 200 | 200 | 200 |

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- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

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| **Project EBIT** | | **40** | **40** | **40** | **40** | **40** |

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| Revenues − COGS − SG&A expenses | | 240 | 240 | 240 | 240 | 240 |
| − Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project EBIT** | | **40** | **40** | **40** | **40** | **40** |
| − Taxes | | 8.4 | 8.4 | 8.4 | 8.4 | 8.4 |

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| **Project EBIT** | | **40** | **40** | **40** | **40** | **40** |
| − Taxes | | 8.4 | 8.4 | 8.4 | 8.4 | 8.4 |
| **Project net income** | | **31.6** | **31.6** | **31.6** | **31.6** | **31.6** |

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| − Taxes | | 8.4 | 8.4 | 8.4 | 8.4 | 8.4 |
| **Project net income** | | **31.6** | **31.6** | **31.6** | **31.6** | **31.6** |
| + Depreciation | | 200 | 200 | 200 | 200 | 200 |

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| **Project EBIT** | | **40** | **40** | **40** | **40** | **40** |
| − Taxes | | 8.4 | 8.4 | 8.4 | 8.4 | 8.4 |
| **Project net income** | | **31.6** | **31.6** | **31.6** | **31.6** | **31.6** |
| + Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project net income + Depreciation** | | **231.6** | **231.6** | **231.6** | **231.6** | **231.6** |

# Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

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| **Project net income** | | **31.6** | **31.6** | **31.6** | **31.6** | **31.6** |
| + Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project net income + Depreciation** | | **231.6** | **231.6** | **231.6** | **231.6** | **231.6** |
| Cash flow from changes in NWC | −100 | 0 | 0 | 0 | 0 | +100 |

# Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
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| − Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project EBIT** | | **40** | **40** | **40** | **40** | **40** |
| − Taxes | | 8.4 | 8.4 | 8.4 | 8.4 | 8.4 |
| **Project net income** | | **31.6** | **31.6** | **31.6** | **31.6** | **31.6** |
| + Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project net income + Depreciation** | | **231.6** | **231.6** | **231.6** | **231.6** | **231.6** |
| Cash flow from changes in NWC | −100 | 0 | 0 | 0 | 0 | +100 |
| Cash flow from capital investments | −1,000 | 0 | 0 | 0 | 0 | 0 |

# Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
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| − Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project EBIT** | | **40** | **40** | **40** | **40** | **40** |
| − Taxes | | 8.4 | 8.4 | 8.4 | 8.4 | 8.4 |
| **Project net income** | | **31.6** | **31.6** | **31.6** | **31.6** | **31.6** |
| + Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project net income + Depreciation** | | **231.6** | **231.6** | **231.6** | **231.6** | **231.6** |
| Cash flow from changes in NWC | −100 | 0 | 0 | 0 | 0 | +100 |
| Cash flow from capital investments | −1,000 | 0 | 0 | 0 | 0 | 0 |
| **Total cash flows** | **−1,100** | **231.6** | **231.6** | **231.6** | **231.6** | **331.6** |

# Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Revenues − COGS − SG&A expenses | | 240 | 240 | 240 | 240 | 240 |
| − Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project EBIT** | | **40** | **40** | **40** | **40** | **40** |
| − Taxes | | 8.4 | 8.4 | 8.4 | 8.4 | 8.4 |
| **Project net income** | | **31.6** | **31.6** | **31.6** | **31.6** | **31.6** |
| + Depreciation | | 200 | 200 | 200 | 200 | 200 |
| **Project net income + Depreciation** | | **231.6** | **231.6** | **231.6** | **231.6** | **231.6** |
| Cash flow from changes in NWC | −100 | 0 | 0 | 0 | 0 | +100 |
| Cash flow from capital investments | −1,000 | 0 | 0 | 0 | 0 | 0 |
| **Total cash flows** | **−1,100** | **231.6** | **231.6** | **231.6** | **231.6** | **331.6** |
| **NPV** | **−107.23** | | | | | |

- NPV = −$107.23 million, reject the project

# Scenario Analysis

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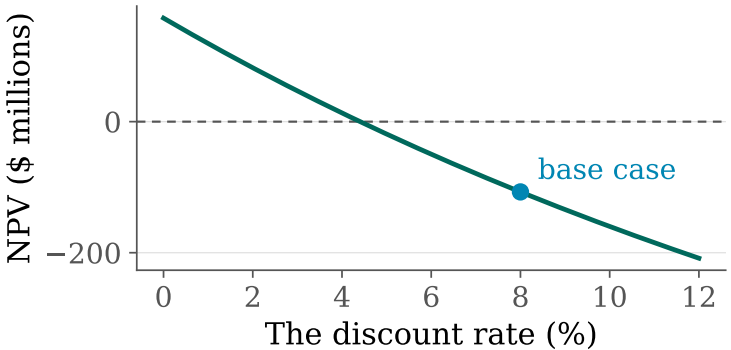
- The inputs behind a project's cash flows are forecasts, and forecasts can be wrong
- **Scenario analysis**, recompute the NPV under different assumptions about the inputs
- E.g., for the data center, how does the NPV change with the discount rate, with what it brings in a year, or with the investment in fixed assets?

# Example, Scenario Analysis

- How does the NPV change with the discount rate?

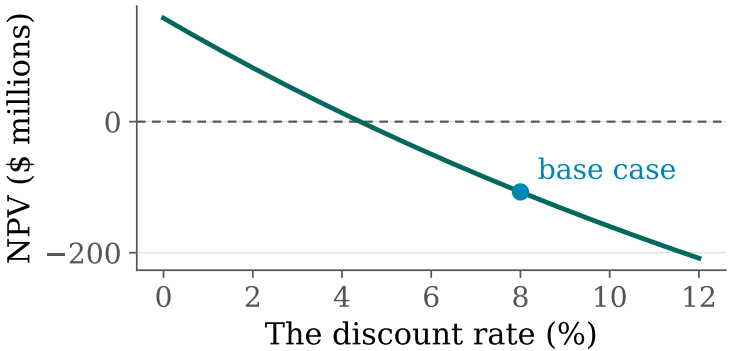
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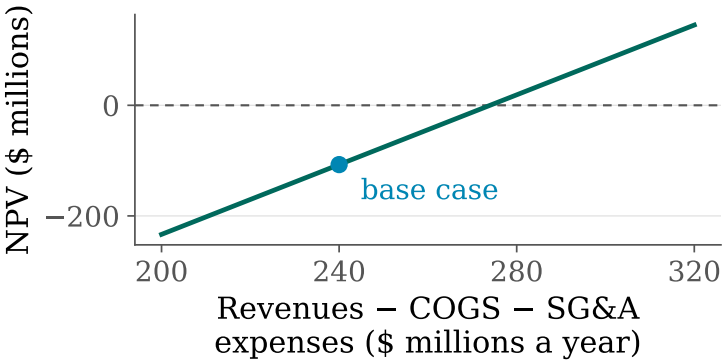

- NPV falls as the discount rate rises, positive only below the IRR of 4.41%

# Example, Scenario Analysis

- How does the NPV change with what the data center brings in a year?

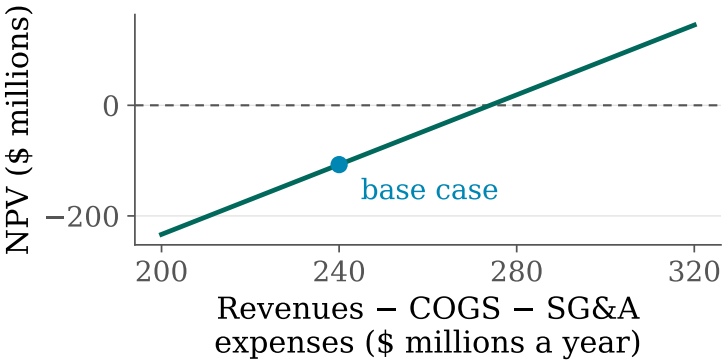
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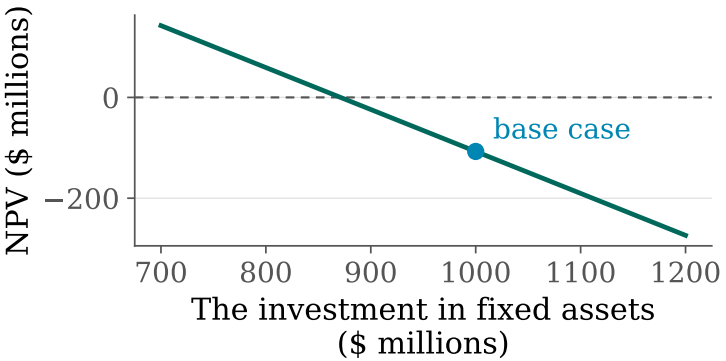

- The data center must bring in about 274 a year after COGS and SG&A expenses for the NPV to turn positive

# Example, Scenario Analysis

- How does the NPV change with the investment in fixed assets?

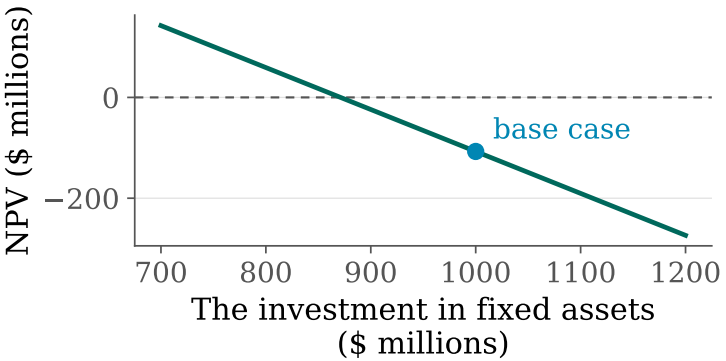
# Example, Scenario Analysis

- How does the NPV change with the investment in fixed assets?



# Example, Scenario Analysis

- How does the NPV change with the investment in fixed assets?




- If the data center keeps bringing in 240 a year after COGS and SG&A expenses, the investment in fixed assets must come in under about 871 for the NPV to turn positive

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- In-class demo, I give the AI a prompt and run the code it returns

## Module 7, Part 5

# Practice Problem

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- The project ends after 6 years. The corporate tax rate is 21%, and the discount rate is 12%
- What is the project's NPV?

# Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |

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| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| AI chips sold (thousands) | | 1,000 | 1,050 | 1,103 | 1,158 | 1,216 | 1,276 |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |

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- Depreciation is not a cash flow, add it back ($ thousands),

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
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| **Project net income + Depreciation** | | **1,790** | **1,869** | **1,952** | **2,039** | **2,130** | **2,227** |

- The cash for the machinery went out at $t = 0$, it is counted in Component 4

# Solution, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 250 while it operates, recovered at $t = 6$ ($ thousands),

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

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- The project's NWC is 250 while it operates, recovered at $t = 6$ ($ thousands),

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| The project's NWC | 250 | 250 | 250 | 250 | 250 | 250 | 0 |

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| Change in NWC | +250 | 0 | 0 | 0 | 0 | 0 | −250 |
| **Cash flow from changes in NWC** | **−250** | **0** | **0** | **0** | **0** | **0** | **+250** |

- An increase in NWC is a negative cash flow, and a decrease is a positive cash flow

# Solution, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets (the machinery) at $t = 0$ ($ thousands),

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

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|---|---|---|---|---|---|---|---|
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# Solution, Component 4, Cash Flow from Capital Investments

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|---|---|---|---|---|---|---|---|
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| **Cash flow from capital investments** | **−6,000** | **0** | **0** | **0** | **0** | **0** | **0** |

# Solution, Total Cash Flows

- Add up the four components year by year ($ thousands),

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|---|---|---|---|---|---|---|---|
| | | | | | | | |

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| **Total cash flows** | **−6,250** | **1,790** | **1,869** | **1,952** | **2,039** | **2,130** | **2,477** |

- At $t = 0$, $-6{,}000 - 250 = -6{,}250$. At $t = 6$, $2{,}227 + 250 = 2{,}477$

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- Discount the total cash flows at the 12% discount rate ($ thousands),

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- NPV $> 0$, so the company should launch the new design

# Key Takeaways

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- Discount cash flows, not net income, and exclude cash flows from financing activities
- Only incremental cash flows count, include indirect effects, opportunity costs, changes in NWC, and terminal cash flows, exclude sunk costs
- Lay the years out in a table, total the cash flows, discount, and apply the NPV rule

# Thank you!

Please let me know if you have any questions.